

Fig. 3: Actual-size, single-side PCB for the computerised electronic lock

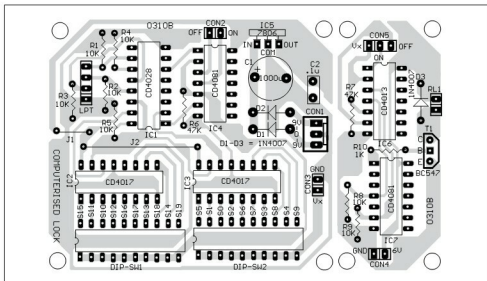


Fig. 4: Component layout for the PCB

allows more sophisticated functionality than mechanical locks.

Electronic locks provide for increased security as the lock can be reprogrammed to reject the keys that it would previously accept. Also, these don't use a mechanical key that can be easily duplicated.

Parallel port

The parallel port or line printer terminal (LPT) is a 25-pin, D-type connector (DB25) at the back of your PC. It comprises three different ports, namely, data port, control port and status port. In order to access any port, you should have its port address. The addresses of data port, status port and control port

are '0x378', '0x378+1' and '0x378+2', respectively. Here we have used two data lines along with signal ground for controlling the direction of motor rotation.

Circuit description

Fig. 1 shows the circuit of the computerised electronic lock. It comprises BCD-to-decimal decoder CD4028, decade counter CD4017, AND gate CD4081, flip-flop CD4013 and a few discrete components.

BCD-to-decimal decoder CD4028 (IC1) consists of four inputs, decoding logic gates and ten output buffers. A BCD code applied to its input pins (D0 through D3) results in a high level at

PARTS LIST

Semiconductors:

IC1	- CD4028 BCD-to-decimal decoder
IC2, IC3	- CD4017 decade counter
IC4, IC6	- CD4081 AND gate
IC5	- 7806, 6V regulator
IC7	- CD4013 flip-flop
T1	- BC547 npn transistor
D1-D3	- 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R5, R8, R9	- 10-kilo-ohm
R6, R7	- 47-kilo-ohm
R10	- 1-kilo-ohm

Capacitors:

C1	- 1000 μ F, 25V electrolytic
C2	- 0.1 μ F ceramic disk

Miscellaneous:

X1	- 230V primary to 9V-0-9V, 500mA secondary transformer
RL1	- 6V, 1C/O relay
DIP-SW1, DIP-SW2	- 8-way DIP switch

the selected 1-of-10 decimal decoded outputs. Pins D0 through D3 of CD4028 are pulled low through resistors R1 through R4 and receive inputs from the parallel port. Data pins D0 through D2 of the parallel port control the locking and unlocking processes.

The CD4017 is a 5-stage divide-by-10 counter with ten decoded outputs and a carry-out bit. These counters are cleared to their zero count by a high pulse on their reset pin 15, which is provided by Q1 output of decoder CD4028.

Schmitt trigger action in the clock input circuit provides pulse that allows unlimited clock input pulse rise and fall times. Q2 and Q3 outputs of CD4028 provide positive-edge clock pulses to IC3 and IC2, respectively. Both of these counters advance on the positive edge of the clock signal when strobe pin 13 is in logical '0' state. Q4 and Q5 outputs of CD4028 control the lock 'on' and 'off' functions with the help of AND gate CD4081.

Q0 through Q9 outputs of decade counters CD4017 are used for selection of the device with the help of DIP switches DIP-SW1 and DIP-SW2. Q0 through Q9 outputs of IC2 are used for device selection in decade from 10 through 90 (10, 20, 30, ..., 70, 80, 90),